Comprehensive Evaluation Plan: SAGE-SRL

This document outlines the full evaluation strategy for the SAGE-SRL architecture. Expanding upon prior security and pedagogical evaluations, this strategy introduces four distinct evaluation tracks to rigorously validate the system's novel Mastery Tracking, Self-Regulated Learning (SRL), Misconception Detection, and Prerequisite Enforcement capabilities.

Track 1: Software-Defined Testing (LLM-as-Judge)

Methodology: Scripted interactions through the actual SAGE-SRL system, evaluated by an LLM-as-Judge using strict rubrics.

1.1 Mastery-Aware Adaptation Quality

Objective: Test whether the system appropriately adapts its pedagogical responses based on the student's Bayesian Knowledge Tracing (BKT) state.

- **Method:** Ask the same concept question simulating three distinct student states:
    1. Fresh student (no mastery history)
    2. Student with quiz-only mastery ($\tilde{P}_{quiz} = 0.95$, others at prior)
    3. Fully certified student reviewing after knowledge decay
- **LLM Judge Rubric (1-5 scale):** Appropriate difficulty, scaffolding depth, and assumed prerequisite knowledge.
- **Dataset:** 20 C concepts $\times$ 3 student states = 60 interactions.

1.2 Prerequisite Gate Enforcement

Objective: Validate the strictness and contextual awareness of the prerequisite gating system.

- **Method:** Simulate access attempts under varying conditions:
    1. Attempting to access an advanced topic (e.g., pointers) without mastering the prerequisite (e.g., variables).
    2. A previously certified student who decayed below $\theta_{decertify}$ attempting to access a dependent topic (should pass via ever_certified flag).
    3. A student who explicitly failed/never mastered the prerequisite trying to skip ahead.
- **LLM Judge Rubric:** Correctness of block/allow decision, appropriateness of the explanation/redirection.
- **Dataset:** 15 prerequisite pairs $\times$ 3 scenarios = 45 interactions.

1.3 Misconception Detection & Remediation

Objective: Evaluate the Exponential Moving Average (EMA) misconception tracker and the system's targeted remediation capabilities.

- **Method:** Simulate a student exhibiting specific C programming misconceptions (e.g., "arrays start at index 1", "== is assignment").
- **LLM Judge Rubric (1-5 scale):** Detection accuracy (within 2 interactions), remediation quality (targeted vs. generic), and verification of resolution.
- **Dataset:** 10 common C misconceptions $\times$ 3 severity levels = 30 interaction chains.

1.4 Verification Quiz Quality

Objective: Assess the dynamic generation and assessment quality of the explicit verification flow.

- **Method:** Trigger the "I know X (verify)" flow for various concepts.
- **LLM Judge Rubric (1-5 scale):** Question relevance, conceptual diversity, appropriate difficulty spread, and fairness of the grading logic.
- **Dataset:** 15 concepts $\times$ verification flow = 15 multi-turn sessions.

Track 2: Comparative Evaluation

Methodology: Direct comparison against baseline LLMs and a feature analysis against existing literature to establish state-of-the-art positioning.

2.1 5-Way Mastery & SRL Comparison

Objective: Establish the headline superiority of SAGE-SRL's architecture over raw and prompted LLMs.

- **Baselines:**
  1. GPT-4o (Raw) - No tutor prompt, no BKT, no SRL
  2. GPT-4o (Tutor Prompt) - System prompt only, no BKT, no SRL
  3. Gemini (Raw) - No tutor prompt, no BKT, no SRL
  4. Gemini (Tutor Prompt) - System prompt only, no BKT, no SRL
  5. **SAGE-SRL (Ours)** - Full system with BKT + SRL
- **Metrics:** Mastery Sensitivity, Prerequisite Awareness, Misconception Handling, Review Prompting, Scaffolding Persistence, Code Density (carry-over), Security Failure Rate (carry-over).
- **Dataset:** 40 queries (10 Concept, 10 Problem, 10 Mastery-probe, 10 Review-trigger).

2.2 System Feature Comparison (Literature)

Objective: Position SAGE-SRL against systems that cannot be directly queried via API.

- **Comparison Matrix:**

Feature	SAGE-SR	Khanmig	ALEKS	MATHia	CodeHel	Korbit

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LLM-based responses	Yes	Yes	No	No	Yes	Yes
Multi-tier BKT	Yes	No	No	Yes*	No	No
Conjunctive mastery	Yes	No	No	No	No	No
Forgetting model	Yes	Unknown	Yes	No	No	Unknown
SM-2 spaced repetition	Yes	No	Yes	No	No	Unknown
Misconception EMA	Yes	No	No	Yes*	No	No
Prerequisite gates	Yes	Partial	Yes	Yes	No	Partial
Adaptive calibration	Yes	No	Unknown	No	No	No
Security guardrails	Yes	Yes	N/A	N/A	Partial	Partial
Domain	C Prog.	Math/Gen	Math	Math	CS	STEM
* *Indicates*						

a different implementation approach.						

Track 3: Ablation Study

Methodology: Isolate and quantify the contribution of individual architectural components to prove their necessity.

Dataset: 40-query dataset from Track 2.1, scored via LLM-as-Judge, plus synthetic metrics (False-mastery rate, F1 score, Retention window).

Variant	**BKT**	**SRL (SM-2 + Decay)**	**Misconception**	**Prereq Gate**	**Calibrator**	**Verification**
Full System	Factored	Yes	Yes	Yes (ever_certified)	Adaptive	3-question
— **BKT**	None (pooled to single score)	Yes	Yes	Row-existence only	Fixed	1-question
— **SRL**	Factored	No decay, no SM-2	Yes	Yes	Adaptive	3-question
— **Misconception**	Factored	Yes	Disabled	Yes	Adaptive	3-question
— **Prerequisites**	Factored	Yes	Yes	Disabled	Adaptive	3-question
—	Factored	Yes	Yes	Yes	Fixed	3-question

Calibrator					$\theta =$	
Bare LLM	None	None	None	None	None	None

Track 4: Synthetic BKT Studies

Methodology: Mathematical and statistical simulations validating the theoretical underpinnings of the knowledge tracing logic.

- **Studies 1-3:** Factored BKT vs. Evidence Dilution Problem (EDP)
- **Study 4-5:** Adaptive Decay & Retention/Forgetting curves
- **Study 6:** Misconception EMA analytical convergence
- **Study 7:** Verification Flow probability matching (P_G^3 bounds)
- **Study 8:** End-to-End SRL Loop simulation over time

Summary: Full Evaluation Map

#	Evaluation Type	What it Proves	Proposed Figures / Tables
1.1	Mastery-Aware Adaptation	System adapts dynamically to student state	Table + Radar chart
1.2	Prerequisite Gate	Gating logic functions correctly (inc. ever_certified)	Table (Pass/Fail matrix)
1.3	Misconception Detection	EMA detects and effectively remediates errors	Table + Example conversation chains
1.4	Verification Quiz Quality	3-question flow produces valid, diverse questions	Table
2.1	5-Way Mastery Comparison	SAGE-SRL outperforms baselines on mastery/SRL	Bar chart + Win-rate table

2.2	Feature Comparison	SAGE-SRL pushes the state-of-the-art vs literature	Feature matrix table
3.0	Ablation Study	Every component contributes to the final metrics	Ablation table (7 variants $\times$ 8 metrics)
4.1-3	BKT vs EDP Studies	Factored BKT solves the Evidence Dilution Problem	3 analytical figures
4.5	Forgetting/Retention	Adaptive decay appropriately models memory loss	Decay curves figure
4.6	Misconception EMA	EMA tracking converges mathematically as claimed	Trajectory figure
4.7	Verification Flow	Verification probabilities match mathematical bounds	Bar chart
4.8	End-to-End SRL Loop	The full learning cycle functions over a timespan	Learning journey timeline figure